Pranati Modumudi

https://github.com/pranmod01 | (408)-464-8674 | pranati.m@columbia.edu | linkedin.com/in/pranatimodumudi/

EDUCATION

Columbia University Aug 2025 - May 2027 (Expected)

M.S. in Computer Science, MS Thesis Track (Machine Learning and Computational Neuroscience)

Relevant Coursework: LLM-Based Generative AI, Ethical AI, LLM Interpretability and Alignment, Computation and the Brain, Theoretical Neuroscience

University of California, Berkeley Aug 2019 - May 2023

B.A. Data Science Honors and B.A. Economics | Regents and Chancellor's Scholar | Berkeley SCET

Relevant Coursework: Linear Algebra and Differential Equations; Probability & Probabilistic Modeling; Data Engineering; NLP; Game Theory

SKILLS

- **ML/DL & RL**: PyTorch, JAX, Scikit-learn, Neural Networks, Reinforcement Learning, Model Training & Evaluation, Distributed Training, MLflow, Weights & Biases
- **LLMs & Generative AI:** Large Language Models, Prompt Engineering, RAG, RLHF, Fine-Tuning, MCP, Langchain, HuggingFace, AllenNLP, SpaCy
- **Programming & Systems:** Python, R, SQL, Bash, Java, Git, Docker, Databricks, Snowflake, Big Data Systems, ETL, APIs
- **Neuroscience & Signal Processing:** EEG, Eye-Tracking, Neural Data Analysis, Real-Time Signal Processing, BCIs

RESEARCH

MS Thesis & Research | Zemel Group, Columbia University

- Designing a brain-inspired continual learning framework drawing on hippocampal replay and memory consolidation to address catastrophic forgetting in neural networks.
- Co-first author on CuedMemVLM, a VLM that compresses visual inputs into memory tokens for long-sequence reasoning, benchmarked against human memory behaviors (in preparation for ICLR submission).

Human Decision Modeling | Laboratory for Intelligent Imaging and Neural Computing, Columbia University

- Developed a VR value-based decision task (3 ambiguity levels; EEG, pupillometry, eye-gaze, behavioral data; N=57) using weighted late-fusion classifiers (74.9% accuracy, LOSO CV).
- Found physiological signals contributed minimally within-session but were critical for cross-session generalization (+2.6pp, p=0.008), with implications for reducing recalibration in physiological BCIs (manuscript under review, SfN 2026).

Undergraduate Senior Honors Thesis | Redwood Center for Theoretical Neuroscience, UC Berkeley

- Modeled hippocampal place cell representations across species; advised by Prof. Friedrich Sommer & Dr. Gautam Agarwal.

INDUSTRY

Machine Learning Engineer Intern, Twitch, San Francisco June 2026 - Sept 2026

Data Scientist I, Disney Streaming, New York Aug 2023 - May 2025

- Built production ML pipelines (Bayesian structural time-series, gradient boosted decision trees) for causal inference on user behavior, serving 150M+ subscribers with <100ms real-time prediction latency.

Machine Learning Systems Intern, OctoML (acquired by Nvidia), Remote Oct 2022 - Dec 2022

- Optimized the TVM ML compiler through 100+ tuning experiments, reducing convergence time by 10%; findings published in the MLArchSys International Symposium on Computer Architecture 2022.

SELECTED PROJECTS

Motor Intent Detection in Hybrid BCIs | Personal Project

- Built an EEG classification pipeline using CSP+LDA, EEGNet, and LSTM to distinguish volitional motor intent from externally-evoked responses, improving specificity from 51% to 83% for rehabilitation brain-computer interfaces.

Sycophancy Robustness in Multi-Agent LLM Systems | Apart Research AI Manipulation Hackathon

- Investigated sycophancy robustness in multi-agent LLM systems across 120 experiments; found adversarial agent pressure breaks anti-sycophancy defenses at a 70% flip rate, with accuracy inversely correlated with social pressure.

Multi-Agent SLM Evaluators for AI Slop Detection | AWS Small Language Build Day Hackathon

- Built an ensemble of four specialized small language models (1-3B parameters) fine-tuned with LoRA on AWS Trainium to detect low-fidelity model outputs, enabling order-of-magnitude cost reduction vs. large-model inference for content quality monitoring.